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CO2 -AT1

MLA0201-Fundamentals of Machine Learning

QUESTION 1: Linear Regression Model

House	Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct the Linear Regression Model

The model is

$$Y = \beta_0 + \beta_1(\text{Area}) + \beta_2(\text{Bedrooms})$$

where

- Y = House Price
- β_0 = Intercept
- β_1 = Area coefficient
- β_2 = Bedroom coefficient

(ii) Estimate the Regression Coefficients

Using the given data (approximate values),

$$\begin{aligned}\beta_0 &= 5 \\ \beta_1 &= 0.04 \\ \beta_2 &= 2\end{aligned}$$

Therefore,

$$Y = 5 + 0.04(\text{Area}) + 2(\text{Bedrooms})$$

(iii) Interpretation

- **Intercept ($\beta_0 = 5$):** Base house price is **₹5 Lakhs**.
- **Area coefficient ($\beta_1 = 0.04$):** Every additional **1 sq.ft** increases the price by **₹0.04 Lakhs**.
- **Bedroom coefficient ($\beta_2 = 2$):** Every additional bedroom increases the price by **₹2 Lakhs**.

Final Answer

Regression Equation:

$$Y = 5 + 0.04(\text{Area}) + 2(\text{Bedrooms})$$

Q2. Linear Classification

(i) Construct the Linear Classification Model

Assume

$$f(x) = w_1(\text{Offer}) + w_2(\text{Win}) + b$$

Take

$$w_1 = 1, w_2 = 1, b = -3$$

Hence,

$$f(x) = \text{Offer} + \text{Win} - 3$$

(ii) Decision Boundary

Decision boundary:

$$Offer + Win - 3 = 0$$

or

$$Offer + Win = 3$$

Classification Rule:

- If $Offer + Win > 3 \rightarrow$ Spam (1)
- If $Offer + Win \leq 3 \rightarrow$ Not Spam (0)

(iii) Classify New Email

Given

Offer = 2

Win = 1

$$f(x) = 2 + 1 - 3 = 0$$

Since the value is on the decision boundary,

Predicted Class = Spam (1)

Final Answer

Linear Model:

$$f(x) = Offer + Win - 3$$

Decision Boundary:

$$Offer + Win = 3$$

Prediction:

Spam

Q3. Naïve Bayes Classification

Patient	Fever	Headache	Disease
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Prior Probabilities

Disease = Yes = 3

Disease = No = 2

Total = 5

$$P(Yes) = \frac{3}{5} = 0.6$$

$$P(No) = \frac{2}{5} = 0.4$$

(ii) Conditional Probabilities

For Disease = Yes

There are 3 records.

$$P(Fever = 1 \mid Yes) = \frac{3}{3} = 1$$

$$P(Headache = 0 \mid Yes) = \frac{1}{3}$$

For Disease = No

There are 2 records.

$$P(Fever = 1 \mid No) = \frac{0}{2} = 0$$

$$P(Headache = 0 \mid No) = \frac{1}{2}$$

(iii) Probability for

Fever = 1

Headache = 0

Disease = Yes

$$\begin{aligned} &P(Yes) \times P(Fever = 1 \mid Yes) \times P(Headache = 0 \mid Yes) \\ &= 0.6 \times 1 \times \frac{1}{3} \\ &= 0.2 \end{aligned}$$

Disease = No

$$\begin{aligned} &P(No) \times P(Fever = 1 \mid No) \times P(Headache = 0 \mid No) \\ &= 0.4 \times 0 \times \frac{1}{2} \\ &= 0 \end{aligned}$$

Since

$$0.2 > 0$$

the patient is classified as Disease = Yes.

Final Answer

Prior Probabilities:

$$P(Yes) = 0.6, P(No) = 0.4$$

Conditional Probabilities:

$$\begin{aligned} P(Fever = 1 \mid Yes) &= 1 \\ P(Headache = 0 \mid Yes) &= \frac{1}{3} \\ P(Fever = 1 \mid No) &= 0 \\ P(Headache = 0 \mid No) &= \frac{1}{2} \end{aligned}$$

Prediction:

Disease = Yes

Q4. Linear Discriminant Function (5 Marks)

Given Dataset

Data Point	X_1	X_2	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct the Discriminant Function

A simple linear discriminant function is

$$g(x) = X_1 + X_2 - 9$$

where

- $g(x) > 0 \rightarrow \text{Class C2}$
- $g(x) \leq 0 \rightarrow \text{Class C1}$

(ii) Decision Boundary

The decision boundary is obtained by setting

$$g(x) = 0$$

Therefore,

$$X_1 + X_2 - 9 = 0$$

or

$$\boxed{X_1 + X_2 = 9}$$

(iii) Classify the New Point

Given

$$X_1 = 3, X_2 = 3$$

Substitute into the discriminant function:

$$g(x) = 3 + 3 - 9 = -3$$

Since

$$g(x) < 0$$

the point belongs to

Class C1

Final Answer

Discriminant Function:

$g(x) = X_1 + X_2 - 9$

Decision Boundary:

$X_1 + X_2 = 9$

Predicted Class:

C1

Q5. Bayesian Logistic Regression (5 Marks)

Given Dataset

Customer	Monthly Charges	Tenure	Churn
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

(i) Bayesian Logistic Regression Model

The logistic regression model is

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where

$$z = \beta_0 + \beta_1(\text{Monthly Charges}) + \beta_2(\text{Tenure})$$

(ii) Estimate the Parameters

Approximate coefficients:

$$\begin{aligned}\beta_0 &= -8 \\ \beta_1 &= 0.01 \\ \beta_2 &= -0.20\end{aligned}$$

Hence,

$$z = -8 + 0.01(\text{Monthly Charges}) - 0.20(\text{Tenure})$$

(iii) Predict for

Monthly Charges = 650

Tenure = 9

Substitute values:

$$\begin{aligned}z &= -8 + 0.01(650) - 0.20(9) \\ &= -8 + 6.5 - 1.8 \\ &= -3.3\end{aligned}$$

Probability

$$\begin{aligned}P &= \frac{1}{1 + e^{3.3}} \\ P &\approx 0.036\end{aligned}$$

Since

$$P < 0.5$$

Prediction:

Customer will NOT Churn (0)

Final Answer

Model:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where

$$z = -8 + 0.01(\text{Monthly Charges}) - 0.20(\text{Tenure})$$

Prediction:

No Churn (0)

Q6. K-Nearest Neighbour (KNN) (5 Marks)

Given Dataset

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

New Customer:

Age = 32

Income = 50

k = 3

(i) Compute Euclidean Distance

Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance from C1

$$\begin{aligned}
&= \sqrt{(32 - 25)^2 + (50 - 40)^2} \\
&= \sqrt{49 + 100} \\
&= \sqrt{149} \\
&= \mathbf{12.21}
\end{aligned}$$

Distance from C2

$$\begin{aligned}
&= \sqrt{(32 - 30)^2 + (50 - 60)^2} \\
&= \sqrt{4 + 100} \\
&= \sqrt{104} \\
&= 10.20
\end{aligned}$$

Distance from C3

$$\begin{aligned}
&= \sqrt{(32 - 35)^2 + (50 - 80)^2} \\
&= \sqrt{9 + 900} \\
&= \sqrt{909} \\
&= 30.15
\end{aligned}$$

Distance from C4

$$\begin{aligned}
&= \sqrt{(32 - 40)^2 + (50 - 20)^2} \\
&= \sqrt{64 + 900} \\
&= \sqrt{964} \\
&= 31.05
\end{aligned}$$

(ii) Classify using k = 3

Arrange distances:

Customer	Distance	Class
C2	10.20	A
C1	12.21	A
C3	30.15	B
C4	31.05	B

The nearest 3 neighbours are

- C2 → A
- C1 → A

- $C3 \rightarrow B$

Votes:

- Class A = 2
- Class B = 1

(iii) Final Class Label

Majority vote:

Class A

Final Answer

Distances:

- $C1 = 12.21$
- $C2 = 10.20$
- $C3 = 30.15$
- $C4 = 31.05$

Nearest Neighbours:

C2, C1, C3

Final Prediction:

Class A

Q7. Decision Tree using ID3 Algorithm (5 Marks)

Given Dataset

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) Compute Entropy of Play

There are:

- Yes = 2
- No = 2

Total = 4

Formula:

$$Entropy(S) = -P(Yes)\log_2 P(Yes) - P(No)\log_2 P(No)$$

Substitute values:

$$\begin{aligned} &= -\frac{2}{4}\log_2 \frac{2}{4} - \frac{2}{4}\log_2 \frac{2}{4} \\ &= -0.5(-1) - 0.5(-1) \\ &= 1 \end{aligned}$$

Answer

$Entropy(Play) = 1$

(ii) Information Gain for Outlook

Outlook = Sunny

Records = 2

Play = No, No

Entropy = 0

Outlook = Overcast

Records = 1

Play = Yes

Entropy = 0

Outlook = Rain

Records = 1

Play = Yes

Entropy = 0

Weighted Entropy

$$\begin{aligned} &= \frac{2}{4}(0) + \frac{1}{4}(0) + \frac{1}{4}(0) \\ &= 0 \end{aligned}$$

Information Gain

$$\begin{aligned} IG(Outlook) &= Entropy(S) - Weighted Entropy \\ &= 1 - 0 \\ &= 1 \end{aligned}$$

Answer

$$IG(Outlook) = 1$$

(iii) Root Node

The attribute with the highest Information Gain becomes the root node.

Therefore,

$$\text{Root Node} = \text{Outlook}$$

Final Answer

Entropy:

$$1$$

Information Gain:

$$1$$

Root Node:

$$\text{Outlook}$$

Q8. Linear Separating Hyperplane (5 Marks)

Given Data

X_1 X_2 Class

2 2 +1

4 4 +1

2 4 -1

4 2 -1

(i) Plot the Points

X_2

5 |

4 | •(-1) •(+1)

| (2,4) (4,4)

3 |

2 | •(+1) •(-1)

| (2,2) (4,2)

1 |

1 2 3 4 5

X_1

(ii) Determine a Linear Separating Hyperplane

A simple linear separator is

$$X_2 = X_1$$

Points:

- (2,2) $\rightarrow$ +1

- $(4,4) \rightarrow +1$

lie on the line.

The other two points are on opposite sides.

Thus,

$$\boxed{\text{Hyperplane: } X_2 = X_1}$$

(iii) Decision Boundary

General equation:

$$X_1 - X_2 = 0$$

or

$$\boxed{X_1 - X_2 = 0}$$

Classification Rule:

- If $X_1 - X_2 \geq 0 \rightarrow \text{Class } +1$
- If $X_1 - X_2 < 0 \rightarrow \text{Class } -1$

Final Answer

Hyperplane:

$$\boxed{X_2 = X_1}$$

Decision Boundary:

$$\boxed{X_1 - X_2 = 0}$$

Classification Rule:

- $X_1 - X_2 \geq 0 \rightarrow +1$
- $X_1 - X_2 < 0 \rightarrow -1$

Q9. Confusion Matrix Evaluation (5 Marks)

Given Confusion Matrices

Logistic Regression

Pred + Pred -

Actual + 40 10

Actual - 5 45

Therefore,

- TP = 40
- FN = 10
- FP = 5
- TN = 45

Naïve Bayes

Pred + Pred -

Actual + 35 15

Actual - 10 40

Therefore,

- TP = 35
- FN = 15
- FP = 10
- TN = 40

(i) Compute Accuracy, Precision and Recall

Logistic Regression

Accuracy

Formula

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ &= \frac{40 + 45}{40 + 45 + 5 + 10} \\ &= \frac{85}{100} \\ &= 0.85 \\ \boxed{\text{Accuracy} = 85\%} \end{aligned}$$

Precision

Formula

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{40}{40 + 5} \\ &= \frac{40}{45} \\ &= 0.889 \\ \boxed{\text{Precision} &= 88.9\%} \end{aligned}$$

Recall

Formula

$$\begin{aligned} \text{Recall} &= \frac{TP}{TP + FN} \\ &= \frac{40}{40 + 10} \\ &= \frac{40}{50} \\ &= 0.80 \\ \boxed{\text{Recall} &= 80\%} \end{aligned}$$

Naïve Bayes

Accuracy

$$\begin{aligned} &= \frac{35 + 40}{35 + 40 + 10 + 15} \\ &= \frac{75}{100} \\ &= 0.75 \\ \boxed{\text{Accuracy} &= 75\%} \end{aligned}$$

Precision

$$\begin{aligned} &= \frac{35}{35 + 10} \\ &= \frac{35}{45} \\ &= 0.778 \\ \boxed{\text{Precision} &= 77.8\%} \end{aligned}$$

Recall

$$\begin{aligned} &= \frac{35}{35 + 15} \\ &= \frac{35}{50} \\ &= 0.70 \\ &\boxed{\text{Recall} = 70\%} \end{aligned}$$

(ii) Compare the Models

Metric	Logistic Regression	Naïve Bayes
Accuracy	85%	75%
Precision	88.9%	77.8%
Recall	80%	70%

Final Answer

Since Logistic Regression has higher Accuracy, Precision, and Recall, it performs better.

Logistic Regression is the better model.

Q10. K-Means Clustering (5 Marks)

Given Points

(2, 4), (3, 5), (8, 9), (9, 10)

Initial Centroids

$$\begin{aligned} C_1 &= (2, 4) \\ C_2 &= (8, 9) \end{aligned}$$

(i) Perform One Iteration of K-Means

Distance Calculation

Point	Distance to C_1	Distance to C_2	Nearest Cluster
(2,4)	0	7.81	C_1
(3,5)	1.41	6.40	C_1
(8,9)	7.81	0	C_2
(9,10)	9.22	1.41	C_2

(ii) Assign Clusters

Cluster 1

(2, 4), (3, 5)

Cluster 2

(8, 9), (9, 10)

(iii) Compute New Centroids

Cluster 1

$$\left(\frac{2+3}{2}, \frac{4+5}{2}\right) \\ = (2.5, 4.5)$$

Cluster 2

$$\left(\frac{8+9}{2}, \frac{9+10}{2}\right) \\ = (8.5, 9.5)$$

Final Answer

Cluster Assignment

- Cluster 1: (2,4), (3,5)
- Cluster 2: (8,9), (9,10)

New Centroids

$$C_1 = (2.5, 4.5)$$

$$C_2 = (8.5, 9.5)$$

